

Graphs of parabolas



1 Consider the equation $y = x^2$.

a Complete the table of values:

x	-3	-2	-1	0	1	2	3
y							

b Use the table of values to sketch the graph of the function on a number plane.

c Are the y -values ever negative?

d Find the equation of the axis of symmetry.

e Find the minimum y -value.

f For every y -value greater than 0, how many corresponding x -values are there?

2 Consider the equation $y = -x^2$.

a Complete the table of values:

x	-3	-2	-1	0	1	2	3
y							

b Use the table of values to sketch the graph of the function on a number plane.

c Are the y -values ever positive?

d Find the equation of the axis of symmetry.

e Find the maximum y -value.

3 For each of the following equations:

i Complete the table of values:

x	-3	-2	-1	0	1	2	3
y							

ii Sketch the graph of the function on a number plane.

a $y = 3x^2$

b $y = -2x^2$

c $y = 3x^2 + 3$

d $y = -2x^2 + 5$

e $y = 3x^2 - 3$

f $y = -2x^2 - 2$

g $y = \frac{1}{2}x^2$

h $y = -\frac{1}{2}x^2$

Transformations of parabolas



4 For each of the following equations:

i Complete the table of values:

x	1	2	3	4	5
y					

- ii Sketch the graph of the function on a number plane.
- iii Find the minimum y -value.
- iv Find the x -value that corresponds to the minimum y -value.
- v Find the coordinates of the vertex.

a $y = (x - 3)^2$ **b** $y = -(x - 3)^2$ **c** $y = (x - 1)^2$ **d** $y = (x - 4)^2$

5 For each of the following equations:

i Complete the table of values:

x	-3	-2	-1	0	1
y					

- ii Sketch the graph of the function on a number plane.
- iii Find the minimum y -value.
- iv Find the x -value that corresponds to the minimum y -value.
- v Find the coordinates of the vertex.

a $y = (x + 1)^2$ **b** $y = -(x + 1)^2$ **c** $y = (x + 2)^2$ **d** $y = -(x + 3)^2$

6 For each of the following equations:

- i State the coordinates of the vertex.
- ii Solve for the equation of the axis of symmetry.
- iii Sketch the graph of the function on a number plane.
- iv Plot the axis of symmetry.

a $y = (x - 2)^2$ **b** $y = (x + 3)^2$ **c** $y = (x + 5)^2$ **d** $y = -(x - 5)^2$

7 Describe the transformation required to transform the parabola $y = -x^2$ into the parabola $y = -(x + 4)^2$.

8 State whether the following parabolas will be concave up or concave down:



a $y = (x - 4)^2$

b $y = -(x + 2)^2$

c $y = -(x - 8)^2$

d $y = (x + 7)^2$

9 For each of the following equations:

i Complete the table of values:

x	-2	-1	0	1	2
y					

ii Sketch the graph of the function on a number plane.

iii State the coordinates of the vertex.

a $y = 2x^2$

b $y = 3x^2$

c $y = -4x^2$

d $y = \frac{1}{2}x^2$

10 For each of the following equations:

i Solve for the equation of the axis of symmetry.

ii Sketch the graph of the function and its axis of symmetry on a number plane.

a $y = 4x^2$

b $y = -3x^2$

c $y = \frac{1}{4}x^2$

d $y = -\frac{1}{2}x^2$

11 Describe the transformation required to transform the parabola $y = x^2$ into the parabola $y = 5x^2$.

12 Rewrite the equation, $y = x^2$, after the following transformations take place:

a Vertically translated by 3 units.

b Vertically translated by -2 units.

c Horizontally translated by 5 units.

d Horizontally translated by -4 units.

e Vertically scaled by 2 units.

f Vertically scaled by -3 units.

g Vertically scaled by $\frac{1}{2}$ units.

h Vertically reflected about the x -axis.

13 Graph the equation, $y = x^2$, after the following transformations take place:

a Vertically translated by 4 units.

b Horizontally translated by -5 units.

c Vertically translated about the x axis.

d Vertically scaled by 2 units.



Applications

- 14** William is training for a remote control plane aerobatics competition. He wants to fly the plane along the path of a parabola so he has chosen the equation:

$$y = 3x^2$$

where y is the height in metres of the plane from the ground, and x is the horizontal distance in metres of the plane from its starting point.

- a** Complete the table of values:

x (m)	0	1	2	3	4
y (m)					

- b** Sketch the shape of the path on a number plane.
- c** Find the lowest height of the plane.
- d** Find the x -value that corresponds to the minimum y -value.
- e** Find the coordinates of the vertex.
- 15** On Jupiter the equation, $d = 12.5t^2$, can be used to approximate the distance in metres, d , that an object falls in t seconds, if air resistance is ignored.

- a** Complete the table of values:

time (t)	0	2	4	6
distance (d)				

- b** Sketch the shape of the path on a number plane.
- c** Use the equation to determine the number of seconds, t , that it would take an object to fall 84.4 m. Round the value of t to the nearest second.